RECOVERY AND CONDITIONING IN ARBITRARY-SCALE SUPER-RESOLUTION: A TWO-COORDINATE ANALYSIS

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ABSTRACT

Arbitrary-scale super-resolution networks decode image features at continuous coordinates, often conditioned on target pixel size, or cell. We separate frequency recovery from the response to this condition. Recovery profiles are similar across four tested decoder designs in input-Nyquist-normalised frequency; capacity changes act mainly above a relative frequency of about 0.75. Cell responses cannot be reduced to a fixed low-pass filter acting on recovered energy. On saved DIV2K and Set14 outputs, recovery and fitted filter changes account for about 41% of the measured doubled-cell cost decrease from $\times 2$ to $\times 6$. Non-scalar changes account for much of the remainder and can worsen or correct native errors. A further confound is intervention size: a fixed cell multiplier gives a decoder-input step proportional to inverse scale. Step-normalised responses and matched-step comparisons do not support a universal loss of sensitivity. Existing retraining without cell still loses about 0.1 dB across scales. The resulting audit separates recovery, intervention size and error alignment, with retained negative results and explicit prediction limits.

***Index Terms*—** Arbitrary-scale super-resolution, implicit representation, frequency analysis, scale conditioning

1. INTRODUCTION

LIIF [1] reconstructs images at continuous queries from local features, offsets and cell size. Later designs include local Fourier estimators [2], integrated positional encoding [3], implicit attention [4], neural operators [5], Gaussian splatting [6] and heat-field filtering [7]. An empirical benchmark already attributes much improvement to size and training rather than decoder novelty [8]; a recent spectral decoder trades matched-baseline PSNR for efficiency [9]. We examine what changes in the reconstructed signal.

Frequency explanations have important precedents. HyRA decomposes an SR mapping using its impulse response and shows that nonlinear responses compensate phase distortions of its low-pass component [10]. LIIF itself observed that changing cell at large scales can visibly affect edges without improving PSNR [1]. Integrated encoding and heat fields explicitly relate scale to filtering [3, 7]. We instead hold each LR input and query grid fixed, change only cell, and quantify output change and its alignment with native error. Our non-scalar response is not a nonlinear component in HyRA's sense.

We (i) locate recovery and capacity effects in input-normalised frequency $u = 2sr$, with scale s and output frequency r; (ii) audit a scalar-filter explanation of cell cost, retaining residuals and images that reverse the mean trend; and (iii) distinguish decreasing fixed-multiplier cost from sensitivity to a fixed decoder-input step. These distinctions also separate inference perturbations from the training benefit of cell.

2. SETUP AND FREQUENCY RECOVERY

Models and data. LIIF, LTE and SC-INR (an LTE-derived experimental variant with analytic sinc cell weights) use the EDSR-baseline architecture [11], separately learned encoder weights and a common DIV2K [12] recipe: scales from $\mathcal{U}(1, 4)$, 1000 epochs and three seeds. GaussianSR uses an external 2000-epoch EDSR checkpoint with a different recipe; it is an additional comparison, not a matched ablation. Matched 200-epoch LIIF arms change the encoder to RDN [13] or SwinIR [14], reduce the MLP from 4×256 to 2×64, or remove cell. These are recipe-matched, not parameter- or compute-matched comparisons.

We use anti-aliased bicubic inputs at $\times\{2, 3, 4, 6\}$; $\times 6$ is outside the training range. Crop frequency analyses use 240^2 centre crops and shave four pixels. Define $r = \max(|f_x|, |f_y|)$ in cycles per output pixel, and $u = 2sr$. Recovery is $R_b = 1 - V_b/E_b$, with target energy E_b and native error energy V_b, in bins of width 0.25 in u. Standard full-image Y-PSNR uses Set5, Set14 [15], BSD100 [16], Urban100 [17] and shave equal to scale. Original stages include registered predictions and failures; the residual and step analyses below are retrospective.

What changes recovery. On DIV2K 0869–0900, networks exceed bicubic mainly around and above the input Nyquist limit (Fig. 1a). Cross-scale agreement is approximate, with retained failures in Table 2. Among the three matched decoder families, maximum crop spread is about 0.010 (image-bootstrap interval $[0.007, 0.016]$, excluding GaussianSR). Including GaussianSR, all 54 tested full-image dataset–scale–band comparisons have spread below 0.019. This supports similar tested profiles, not universal decoder equivalence.

Longer training, stronger encoders and a larger MLP

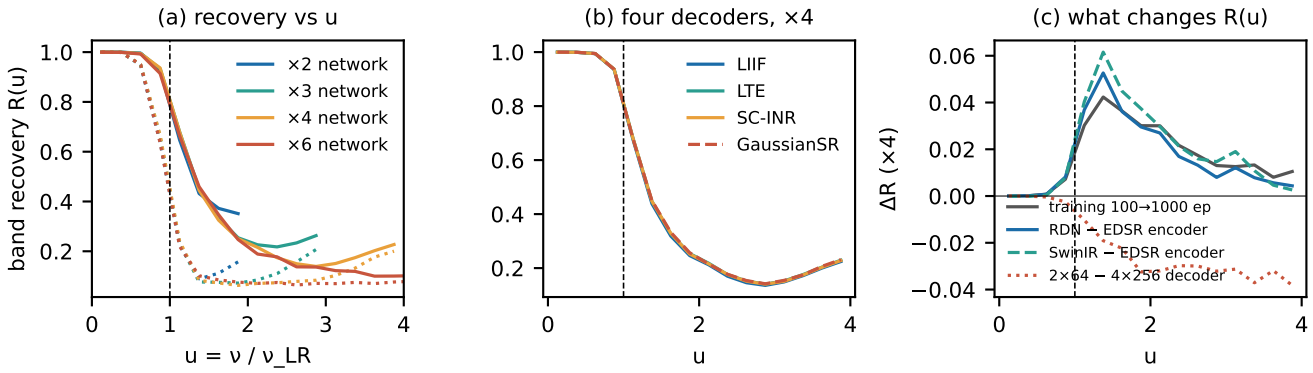


Fig. 1. Recovery in input-normalised frequency u. (a) Three decoder families and three seeds at four scales; dotted curves are bicubic. (b) Decoder comparison at $\times 4$; GaussianSR is an external checkpoint. (c) Effects of training duration, encoder and MLP capacity.

change recovery primarily from $u \approx 0.75$ upward (Fig. 1c). Under Gaussian blur followed by bicubic resizing without antialiasing, a predicted extension of capacity gains to lower frequencies failed: the baseline network already recovers those bands nearly completely. This boundary is empirical, not a theorem about Nyquist. Table 1 reports full-image gains and losses.

3. AUDITING THE CELL RESPONSE

For native output P, perturbed output P_m and target T, write $e = T - P$ and $\delta = P_m - P$. At fixed input and queries, the exact error change is

$$D = \|e - \delta\|^2 - \|e\|^2 = \underbrace{\|\delta\|^2}_{J} - 2\,\mathrm{Re}\langle e, \delta\rangle. \tag{1}$$

Small D may reflect small response, cancellation, or both. All energy terms use RGB-mean MSE units; their shares are not dB shares.

A scalar filter is incomplete. In each output-frequency band fit real $w = \mathrm{Re}\langle P, P_m\rangle / \|P\|^2$ and retain $\epsilon = P_m - wP$. Fit separately by model, scale and multiplier, pooling images and seeds. With $C = \mathrm{Re}\langle e, P\rangle$ and $U = \|P\|^2$, the identity $RE = E - V = U + 2C$ gives

$$\begin{aligned} D &= Q + G + H, \qquad (2)\\ Q &= \sum_b (1 - w_b)^2 R_b E_b,\\ G &= \sum_b 2w_b(1 - w_b)C_b,\\ H &= \sum_b \{\|\epsilon_b\|^2 - 2\,\mathrm{Re}\langle e_b, \epsilon_b\rangle\}. \end{aligned}$$

Fitting makes $\mathrm{Re}\langle P_b, \epsilon_b\rangle = 0$ after pooling; no prediction–error orthogonality is assumed. Image allocations retain $-2(1 - w_b)\,\mathrm{Re}\langle P_b, \epsilon_b\rangle$, which cancels only after pooling.

DIV2K transfer statistics cover 32 images, one seed and 16 bands with $0 < r < 0.5$, excluding DC and the exact Nyquist outer ring. For $m = 2$, Q/D is 0.44, 0.72, 0.81 and 2.50 at $\times 2/3/4/6$. Intermediate-scale agreement partly reflects cancellation between G and H. The fitted strength also changes: in $r \in [0.3125, 0.34375)$, mean w rises from 0.307 to 0.807 between $\times 2$ and $\times 6$. Similar affected frequency ranges do not imply a fixed filter.

Holding HR images and r bands fixed, we symmetrically split ΔQ between changes in $(1 - w)^2$ and RE, averaging the two orders of changing them. Recovery and filter changes account for 22.7% and 18.7% of the measured $\times 2 \to \times 6$ decrease; G, H account for 4.2%,54.4%. On saved Set14 predictions (14 images, two seeds), the shares are 27.3%,14.3%,−0.04%,58.5%. The Set14 H share has paired image-bootstrap interval $[42.0, 67.0]$%, conditional on these trained seeds. This is retrospective accounting, not causal or held-out prediction.

Frequency and image structure. Of the Set14 H decrease, 70.9% lies in $r \in [0.1875, 0.375)$; the same band contributes 78.5% on DIV2K. Comic, monarch and zebra contribute 64.1% of the Set14 net H decrease, while barbara contributes −18.8% (Fig. 2). At $\times 2$, barbara's MSE improves by 616×10^{-6} after doubling cell, with a negative H. Non-scalar changes are not uniformly harmful. Removing any one image and refitting w leaves H at 56.8–61.4% of the total decrease. Per-image-and-seed filters still leave 53.9%, so pooling alone does not close the gap.

What the residual contains. Fit real gains separately for each saved image and seed: (A) per band, (B) per band and RGB channel, (C) per Fourier frequency, or (D) per frequency and channel. At $\times 2$, these retain 93.2%, 93.1%, 45.5% and 40.7% of pooled residual energy; at $\times 6$, 84.7%, 84.3%, 35.4% and 29.0%. Within-band frequency variation matters more than RGB sharing. Even D cannot represent the quadrature component relative to native phase; negative gains absorb

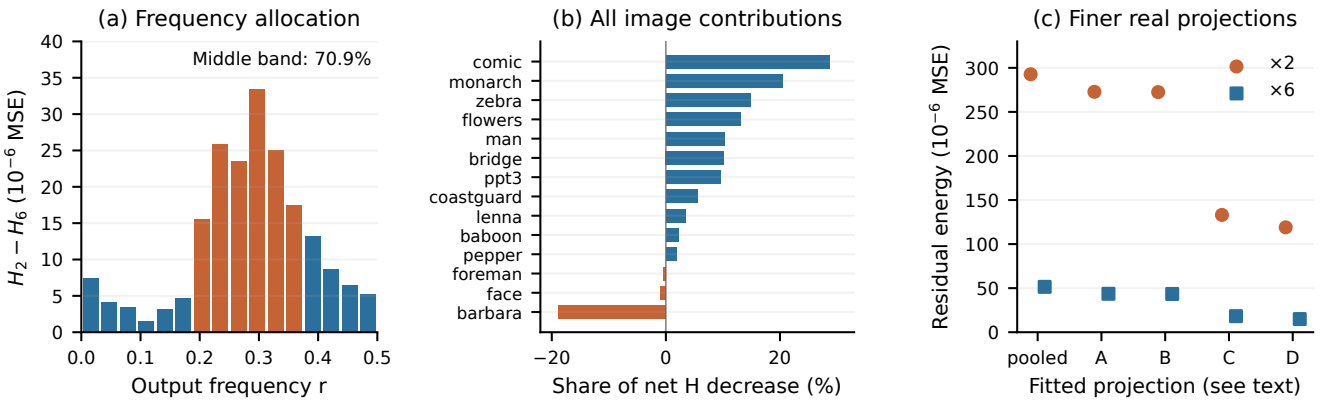


Fig. 2. Saved Set14 outputs, $m = 2$, $0 < r < 0.5$. (a) Frequency allocation of the $\times 2 \to \times 6$ decrease in H. (b) Signed image contributions, retaining all 14 images. (c) Residual energy under finer real projections. Channel and frequency refinements are separate branches; extra fit flexibility is diagnostic, not predictive evidence.

π flips. Its residual energy falls from 119.0 to 14.9 in 10^{-6} MSE, but its contribution to error changes from $+84.8$ to -1.9. These are fitted subspace diagnostics, not an identified network mechanism or evidence of predictive improvement.

4. INTERVENTION SIZE, COST AND TRAINING

Fixed multiplier is not fixed input step. For output side length L, the implementations multiply cell $2/L$ by LR feature-grid size L/s. The decoder therefore receives $c = 2/s$ on each axis, and multiplier m changes it by

$$h = 2(m - 1)/s. \tag{3}$$

Both axes change together; h denotes the per-axis step. For $m = 2$ it shrinks from 1 to $1/3$ between $\times 2$ and $\times 6$. Set14 residual energy decreases from 292.7 to 51.5 in units of 10^{-6} MSE, but division by h^2 gives 292.7, 213.5, 337.9, 463.3 across scales. Raw decline does not establish declining unit-step response. This finite-difference diagnostic is not a Jacobian estimate: features, native cell and relative query coordinates also change with scale.

We reuse original float32 response moments on all 214 Set14/BSD100/Urban100 crops, three models and two seeds. At approximately equal positive step $h = 1/3$, compare $(s, m) = (2, 1.3333)$ and $(6, 2)$; steps differ by 0.01%. Mean output-change energy J at $\times 6$ is 2.26, 1.09 and 0.77 times its $\times 2$ value across these datasets (Fig. 3); paired image-bootstrap intervals are $[1.31, 3.97]$, $[0.91, 1.28]$, $[0.61, 0.96]$. A negative-step control uses $(2, 0.75)$, $(3, 0.625)$, $(4, 0.5)$, all exactly $h = -0.25$, and is likewise dataset-dependent: BSD100 response energy falls 26%, while mean MSE cost rises 218%. These controls challenge a universal sensitivity collapse but do not isolate scale causally: input features and training-support status differ.

Table 1. Matched 200-epoch LIIF arms: mean full-image Y-PSNR change (dB) relative to EDSR LIIF, over four datasets and two seeds. †One seed. Parameter counts and compute differ.

Change	×2	×3	×4	×6
MLP 2×64	-0.25	-0.26	-0.28	-0.24
no cell input	-0.13	-0.09	-0.11	-0.10
RDN encoder	$+0.35$	$+0.29$	$+0.28$	$+0.25$
SwinIR encoder†	$+0.46$	$+0.42$	$+0.42$	$+0.34$

Small cost does not mean no response. On Set14 at $\times 6$, doubled cell changes output by $J = 82.5 \times 10^{-6}$ MSE. Error alignment cancels 62.7×10^{-6}, leaving $D = 19.8 \times 10^{-6}$ (Eq. 1). The dB cost is $10 \log_{10}(1 + D/V)$; native MSE V also grows with scale. The original approximately 2.3/0.40/0.13/0.01 dB sequence on DIV2K remains valid for its fixed-multiplier protocol; it does not establish that cell ceases to act from $\times 3$. Halved-cell and small-perturbation failures remain in the study records.

Training is separate. Removing cell during training costs about 0.1 dB at every benchmark scale, consistently in two seeds (Table 1). A smaller MLP also hurts at every scale, despite the registered prediction that it would be harmless at large scales. Inference perturbations cannot identify why cell improves learning. These results motivate examining trained capacity and conditioning separately, without declaring decoder form generally irrelevant.

5. LIMITS AND CONCLUSION

Evidence concerns the tested models, degradations and finite interventions. Detailed projections use 14 saved Set14 images quantized to float16; source moments, Parseval checks and

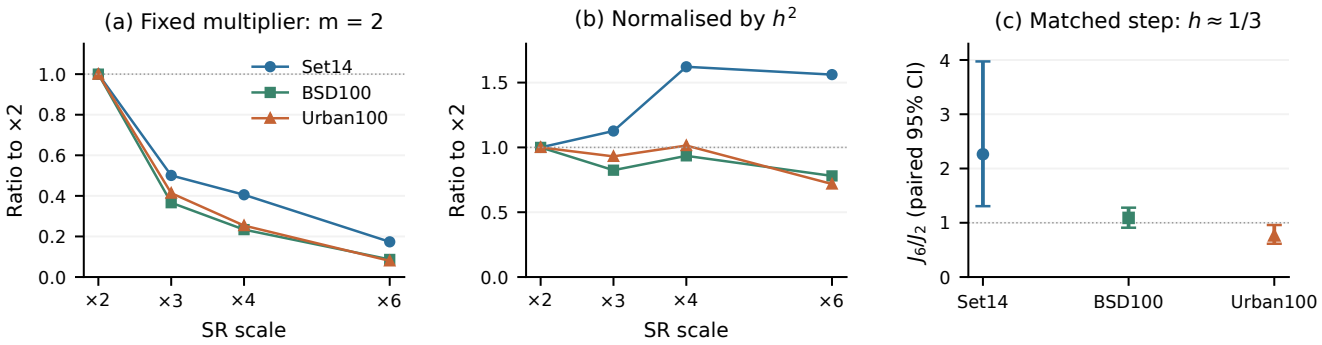


Fig. 3. Original float32 response moments, all non-DC crop frequencies, three models and two seeds. (a) Doubled-cell output-change energy J, relative to $\times 2$. (b) The same comparison after dividing by the squared decoder-input step. (c) Matched positive-step comparison; intervals resample paired images within each dataset, retaining fixed trained seeds.

Table 2. Selected original registered predictions: satisfying cases / total, followed by overall gate status where failed. New residual and step analyses are retrospective.

Prediction	Outcome
strict $R(u)$ collapse (crops; full)	45/54; 53/72 (fail)
decoder spread ≤ 0.03 (crops; full)	36/36; 54/54
doubled-cell peak $r \in [.19, .375)$	36/36
fixed-multiplier cost declines	9/9
$Q/D \in [.7, 1.3]$ at $\times 3/4$	6/6
best cell within 10% of native ($s \leq 4$)	20/27 (fail)
smaller cell costs more	33/144 (fail)
small MLP harmless at large scales	fail
blur moves capacity gains below Nyquist	fail

all-non-DC sensitivity are retained. DIV2K aggregates cannot supply image uncertainty. Bootstrap intervals describe image sampling conditional on existing trained seeds. Image rankings and flexible projections are post hoc, not held-out explanations.

The two frequency coordinates organise recovery and conditioning, but a fixed scalar filter does not explain the measured response. A more complete audit retains non-scalar changes, error alignment and the actual intervention step. It corrects the inference from small net PSNR costs to absent conditioning. Predicting response or benefit from unperturbed outputs, and explaining cell's training benefit, remain open; exact accounting alone is not a new causal theory.

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